Problem 1

a) "Flat wall": the distribution of φ is one-dimensional $\varphi = \varphi(x)$ and obeys the equation:

$$\frac{d^2\varphi}{dx^2} - \frac{\varphi}{a^2} = -\frac{3f}{\varepsilon_0}\delta(x) \qquad (1)$$

Method 1: solution in real space.

When $x \neq a$ $\varphi'' - \varphi/a^2 = 0$, whose solution is

$$\varphi(x) = Ae^{-x/a} + Be^{x/a}$$

As the potential cannot increase away from the wall and has to be continuous at $x=0$:

$$\varphi(x) = Ae^{-\frac{|x|}{a}}$$

The constant A can be found either from the physical condition that near the wall, $x \to 0$, charge of the wall is "unscreened", so that

$$E(x \to 0) = \frac{3f}{2\varepsilon_0} \qquad E(x>0) = -\varphi' = \frac{A}{a}e^{-x/a}$$

so that $$\frac{3f}{2\varepsilon_0} = \frac{A}{a} \qquad (2)$$

Equivalently, one can integrate Eq. (1) over dx from $-\varepsilon$ to $+\varepsilon$ with the subsequent limit of $\varepsilon \to 0$ to be taken. Eq. (2) is recovered then. All together we find

$$\varphi(x) = \frac{3fa}{2\varepsilon_0}e^{-\frac{|x|}{a}} \qquad (3)$$

Method 2: The Fourier transform

$$\left(\varphi(k) = \int_{-\infty}^{\infty} dx\, \varphi(x) e^{-ikx} \right)$$

of Eq. (1) is

$$-k^2 \varphi(k) - \frac{\varphi(k)}{a^2} = -\frac{\rho_f}{\varepsilon_0} \Rightarrow \varphi(k) = \frac{\rho_f/\varepsilon_0}{k^2 + a^{-2}}$$

Inverse Fourier transform:

$$\varphi(x) = \frac{\rho_f}{\varepsilon_0} \int_{-\infty}^{\infty} \frac{dk}{2\pi} \frac{e^{ikx}}{k^2 + a^{-2}}$$

The integral is calculated by the residues method (for $x>0$ the contour is closed in the upper semi-plane, for $x<0$ in the lower), yielding Eq. (3).

B) "Wire". Both methods are equally useful here. Below we use Method 1.

For $r \neq 0$ the equation in the cylindric coordinates is

$$\frac{1}{r}\frac{d}{dr}\left(r \frac{d\varphi}{dr}\right) - \frac{\varphi}{a^2} = 0$$

$$\varphi'' + \frac{1}{r}\varphi' - \frac{\varphi}{a^2} = 0 \qquad (4)$$

This is a particular example of a more general Bessel equation

$$f'' + \frac{1}{\xi} f' + \left(1 - \frac{n^2}{\xi^2}\right) f = 0 \qquad (5)$$

Eq. (5) reduces to Eq. (4) for $n=0$ and $\xi = ir/a$.

The most convenient solutions of the Bessel equation of imaginary argument are modified Bessel funtions I_n and K_n, which are <u>real</u>. The second function, K_n, is also know as Macdonald function:

$$\varphi(r) = A\, I_0(r/a) + B\, K_0(r/a)$$

The function $I_0(ra)$ exponentially grows when $r \to \infty$. Thus, $A = 0$

The remaining constant B can be found from analyzing the $r \to 0$ behavior:

$$K_0(ra \to 0) = -\ln(r/a)$$

This gives for electric field at small distances

$$E(r \to 0) = -\varphi'(r) = \frac{B}{r}$$

On the other hand the field at $r \to 0$ is <u>unscreened</u> and is given by the fiel of a charged wire:

$$E(r) = \frac{\lambda_f}{2\pi\varepsilon_0 r} \qquad (6)$$

This fixes $B = \dfrac{\lambda_f}{2\pi\varepsilon_0}$ and gives the solution at arbitrary

distances

$$\varphi(r) = \frac{\lambda_f}{2\pi\varepsilon_0} K_0(r/a) \qquad (7)$$

The behavior at $ra \gg 1$ is given by the asymptotics of the function K_0:

$$K_0(\xi \gg 1) \approx \sqrt{\frac{\pi}{2\xi}}\, e^{-\xi}$$

which gives

$$\varphi(r \to \infty) \approx \frac{\lambda_f}{2\varepsilon_0} \frac{e^{-r/a}}{\sqrt{2\pi r/a}}$$

<u>Problem 2.</u> The equation for the drift velocities are

$$\frac{d\vec{v}_1}{dt} = -\frac{e}{m_1}\vec{E} - \frac{e}{m_1}\vec{v}_1 \times \vec{B} - \frac{\vec{v}_1}{\tau} = 0,$$

$$\frac{d\vec{v}_2}{dt} = \frac{e}{m_2}\vec{E} + \frac{e}{m_2}\vec{v}_2 \times \vec{B} - \frac{\vec{v}_2}{\tau} = 0,$$

each equation is solved independently, similar to how it was done in class, then $\vec{j} = -en_1\vec{v}_1 + en_2\vec{v}_2$ The total conductivity is the sum of the two conductivities

$$\sigma_{ik} = \sigma_{ik}^{(1)} + \sigma_{ik}^{(2)}$$

where $$\sigma_{ik}^{(1)} = \frac{e^2 n_1 \tau_1}{(\omega_{c1}^2\tau_1^2 + 1)m_1}\begin{pmatrix} 1 & -\omega_{c1}\tau \\ \omega_{c1}\tau & 1 \end{pmatrix}$$

where $\omega_{c1} = \frac{eB}{m_1} > 0$

Similarly,

$$\sigma_{ik}^{(2)} = \frac{e^2 n_2 \tau_2}{(\omega_{c2}^2 \tau^2 + 1) m_2} \begin{pmatrix} 1 & \omega_{c2}\tau \\ -\omega_{c2}\tau & 1 \end{pmatrix} \qquad \omega_{c2} = \frac{eB}{m_2} > 0$$

The Hall current (perpendicular to both $\vec{E}$ and $\vec{B}$) is given by off-diagonal terms

$$j_H = \frac{-e^2 n_1 \tau_1^2 E \omega_{c1}}{(\omega_{c1}^2 \tau^2 + 1) m_1} + \frac{e^2 n_2 \tau_2^2 \omega_{c2} E}{(\omega_{c2}^2 \tau^2 + 1) m_2}$$

For strong magnetic field, such that

$$\omega_{c1}\tau \gg 1 \quad \text{and} \quad \omega_{c2}\tau \gg 1$$

the Hall current simplifies to

$$j_H = -\frac{e^2 n_1}{m_1 \omega_{c1}} E + \frac{e^2 n_2}{m_2 \omega_{c2}} = \frac{e(n_2 - n_1)E}{B}$$

We see that for equal number of electrons and holes (compensated semiconductor) the Hall effect vanishes. Note that for n-type and p-type semiconductors the Hall current has <u>opposite</u> sign.